Time-Aware At-Risk Student Prediction on OULAD

Dual-Cohort Evaluation, Explanation Stability, and Imbalance Robustness

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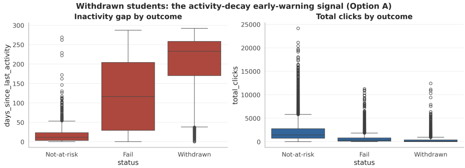
DSP391m Data Science Capstone Project

The problem: failure and dropout in online learning

- Online courses lose many students to **failure** and **withdrawal**.
- Late help does not change the outcome; teachers need to know **who** and **when**, early enough to act.
- And **why** a student is flagged, so the alert is trustworthy and fair.



At-risk students go silent in the VLE well before the deadline.

The catch almost no study checks

- Time-aware models are scored at mid-course on the **full enrolment roster**.
- But that roster still contains students who **already withdrew** — trivial to detect from their inactivity.
- So part of the reported "early-warning" score is really **detecting an outcome that already happened**.

The question this paper asks

Whom are we really predicting, and does the evaluation population inflate the result?

Research questions and contributions

Questions.

RQ1 Which model is best, and *how early* is a prediction reliable (recall and PR-AUC ≥ 0.80)?

RQ2 How *stable* are the explanations (SHAP, LIME)?

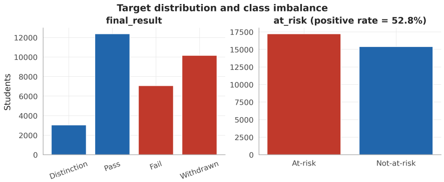
RQ3 Does imbalance handling matter, given at-risk is a slight *majority* (52.8%)?

Contributions.

- **Population-definition inflation, quantified** (gap CI excludes zero at every checkpoint).
- A **verified** time-aware benchmark (Friedman, Wilcoxon–Holm over 25 CV fits).
- Explanation stability judged against a **Monte-Carlo null**, not by eye.
- A **dual-reporting rule** plus an instructor dashboard.

Data: OULAD

- **32,593** enrolments, **28,785** students, **22** module-presentations.
- **7 tables** from **3 systems**: student info, VLE clickstream ($\sim$10.6M rows), assessment.
- Target = {Fail, Withdrawn} = **52.8%** (a slight majority, not a rare class).
- Fixed anonymised snapshot $\Rightarrow$ fully reproducible.



Method: a leakage-safe, time-aware pipeline

Raw (7 tables) → **Frozen student split** → 6 checkpoints (10–100%) → Fit-on-train features → 5 models → SHAP / LIME

- Split **by student**, frozen in git: no student is in both train and test.
- At checkpoint t, only data on or before t enters a feature (no peeking ahead).
- Every transformer, resampler and threshold is fit on the **training fold only**.
- **21 automated tests** guard the pipeline.

The key idea: two questions, two populations

Full cohort (all enrolments)
"Will this end in Fail/Withdrawn?"

- The frame prior work uses.
- Includes students who already left.

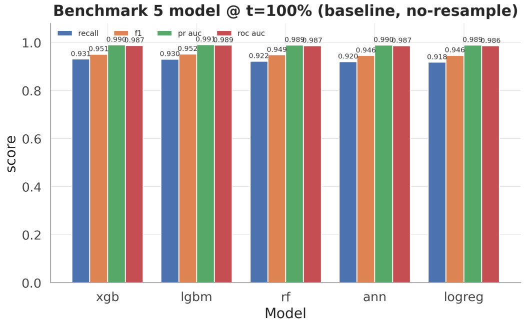
Still-enrolled cohort
"Whom should a tutor contact now?"

- The only students an intervention can reach.
- The real early-warning task.

We score the **same predictions** on both, and report both.

Result 1: which model, and is the ranking real?

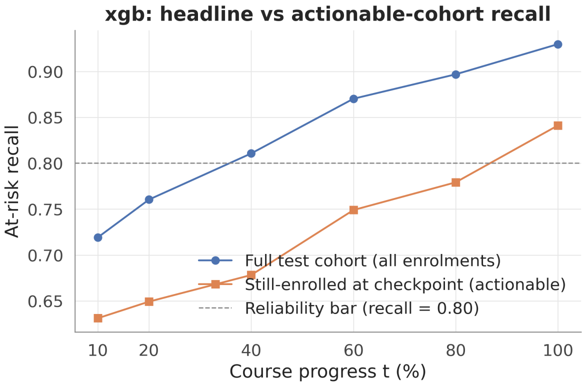
- **XGBoost** leads recall (0.930 at course end); the tree ensembles are close.
- Ranking **verified**: Friedman ($p < 0.05$) + Wilcoxon–Holm over 25 CV fits.
- Logistic regression trails by a few points $\Rightarrow$ the **features** carry the signal.



Result 2 (headline): the population effect

- Full cohort clears recall ≥ 0.80 at $t{=}40\%$ …
- …the **still-enrolled** cohort only at **course end**.
- Gap 95% CI **excludes zero at every checkpoint**.

$t{=}40\%$: 0.811 vs 0.678. $t{=}100\%$: 0.930 vs 0.841.

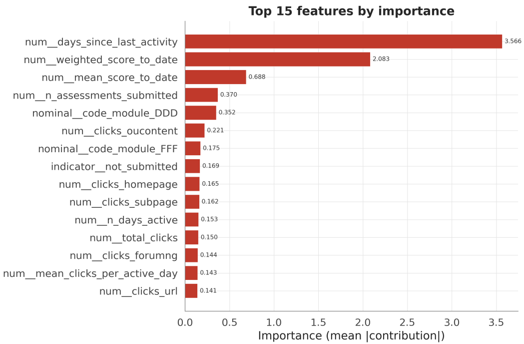


Takeaway

Reliable early warning depends on **which population you score**: 40% of course length on paper, but course end for the students you can actually help.

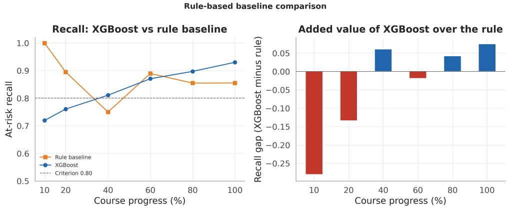
Result 3: explanations — drivers and stability

- Top driver: **days since last VLE activity**, then cumulative score.
- No demographic feature at the top ⇒ alerts are **behaviour-based**, not background-based.
- **Stable**: across seeds, top-10 Jaccard **0.69** vs a Monte-Carlo null of **0.12**.
- SHAP vs LIME agree on the top drivers, diverge in the tail.



Robustness and supporting analyses

- **Imbalance:** across 4 strategies, max Δrecall $= 0.007$ — conclusions do not depend on it.
- **Rule baseline:** a 2-feature rule beats XGBoost early (withdrawn are trivial) but loses late $\Rightarrow$ **hybrid** policy.
- **Calibrated** (ECE 0.018–0.042); **5 features** match the full model; train-test gap shrinks (no overfit).



Limitations (stated openly)

- **Single dataset**; transfer to other institutions is untested.
- **Label conflates** Fail (hard) and Withdrawn (easy, behavioural) — decomposing them is the key next step.
- The top feature is partly a *consequence* of withdrawal, so full-cohort scores read outcomes as much as forecast them; the still-enrolled number (0.841) is the honest one.
- Borderline at t=40%; still-enrolled analysis re-scores a full-roster model.

Conclusion: a costless reporting discipline

Takeaways

- Gradient-boosted trees win a *verified* benchmark; the honest early-warning number is on the **still-enrolled** cohort.
- Explanations are seed-stable and behaviour-based; the top driver *is* the channel of the inflation.

Every early-warning claim should name its cohort, carry its uncertainty, and anchor its explanation to a checkpoint.

Thank you

Questions?

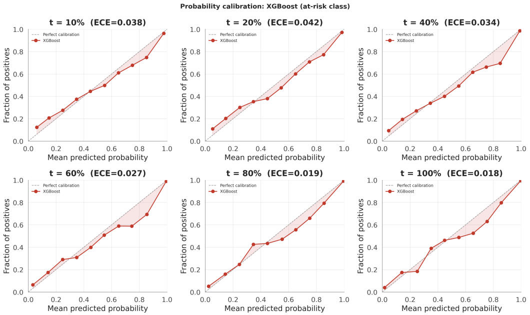
Reproducible pipeline, 21 tests, dashboard, and paper in the repository.
All figures and numbers trace to `reports/tables/*.csv`.

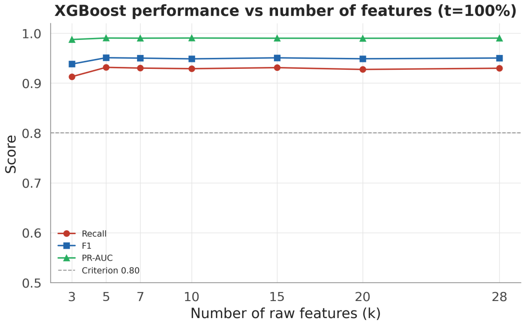
Backup: dual-cohort table

t (%)	Full cohort	Still-enrolled	Gap
40	0.811 [0.798, 0.824]	0.678 [0.656, 0.698]	0.133 [0.122, 0.144]
60	0.870 [0.859, 0.882]	0.749 [0.730, 0.770]	0.121 [0.110, 0.133]
80	0.897 [0.887, 0.907]	0.779 [0.760, 0.800]	0.118 [0.106, 0.130]
100	0.930 [0.921, 0.939]	0.841 [0.821, 0.861]	0.089 [0.076, 0.101]

95% percentile-bootstrap CIs (1,000 student-level resamples).

Backup: calibration and feature ablation





Backup: selected references I

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